Preface to the Second Edition

Near the end of one of the most famous speeches in American history, Abraham Lincoln made reference to "government of the people, by the people, for the people" [5]. In a similar vein, a text about physics and the human body could cover "physics of the body, by the body, and for the body," but perhaps that would be too broad. Our main focus in this second edition is still the physics of the body, namely the physics of physiology, and the text is again largely organized by topics in physiology. There is still a strong emphasis on modeling physiology. As noted in the excellent book by Rideout [7], general modeling in physiology includes that of momentum (such as blood flow), mass (oxygen, carbon dioxide, and so on in the blood), energy (heat flow), and information (signals in nerves, hormones in the bloodstream) [7]. Models coupling multiple systems are also important. (After all, we are all strongly coupled multiple physiological systems, which are, of course, ultimately determined by physics.) We will still confine ourselves to the more physical and less chemical aspects of physiology modeling; chemical modeling is addressed well elsewhere [1, 2, 3, 4, 6, 7]. Within our focus of "physics of the body," we have enhanced the discussion of the physics of disease, aging, motion, and the structure and function of muscles, and now explore how balance, metabolism, and the circulatory system change during pregnancy and birth. The physics of the cross-cutting lifecycle events of pregnancy, injuries, diseases and disorders, aging, and dying transcend the topics of any single chapter. They are now overviewed in the new Appendix F; this appendix also describes how to locate discussions of these topics within the main text.

"Physics by the body" includes the physics of common locomotion and that during athletic events and collisions. These topics were addressed in several parts of the first edition, including many problems. They are now discussed in more detail in both the text and the problems. In particular, the physics of motion, kinesiology, and balance [8] are more directly discussed by the analysis of multisegment modeling and the expanded treatment of the physics of sports. A discussion of human performance of ordinary tasks and at the extremes of athletic competition also transcends the topics of any single chapter and couples many

of them, especially those addressing statics, motion, metabolism, breathing, and blood circulation. To help unify the discussion of athletics, Appendix G has been added to link relevant topics in these chapters and to unify references to different sports activities.

"Physics for the body" includes the physics of measurement analysis and diagnosis and the physics of treatment, much of which is called medical physics. The first edition had some discussion of measurements and diagnosis and essentially none concerning treatment. The presentation of measurement analysis included several methods and a bit about measurement results. A broad overview of this topic is now presented in the new Appendix H, which includes a locator describing where physics analysis of the body is described in the text and problems. Aside from the new brief discussion in Appendix H, treatment is still not addressed in this edition.

This text can still be adapted for use in either lower- or upper-level undergraduate courses, as well as for some low-level graduate courses. It can serve the needs of a primary text for an entire course or as a secondary source, such as through the use of individual chapters. Appendices F, G, and H unite the cross-cutting themes presented in several chapters, and can be used to guide discussion of these topics.

This edition includes numerous additional problems that address the material in the text and explore it in much more detail. Many additional solutions are now provided at the end of the book. The author strongly recommends that you at least peruse the content of the related problems at the end of the chapter when you read any section of the text (even if you do not plan on solving them) because many essential approaches and ideas are covered only in the problems.

Several excellent texts on related topics were cited in the previous edition. In particular, we note again several of the outstanding books describing mathematical models of physiology [1, 2, 3, 4, 6, 7], which are for the most part at a level more advanced than that of this text. Also, of more general interest is the recent "The Story of the Human Body" by Daniel E. Lieberman, which discusses the evolution of the human body and includes extraordinary insights into the structure and function of the bodies of modern-day humans.

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